

Chapter 2.

Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

2.1 Sets

2.2 Set Operations

2.3 Functions

2.4 Sequences and Summations

☆ 2.5 Cardinality of Sets 重点!

2.6 Matrices

Introduction

Sets (集合) are one of the basic building blocks for the types of objects considered in discrete mathematics.

- Important for **counting**.
- Programming languages have **set operations**.

Set theory (集合論) is an important branch of mathematics.

- Many different systems of **axioms have been used to develop set theory**.
- Here we are not concerned with a formal set of axioms for set theory. Instead, we will use what is called naïve set theory.

Sets

A **set** is an unordered collection of objects.

- the students in this class
- the chairs in this room

The objects in a set are called the **elements** (元素), or **members** of the set. A set is said to **contain** (包含) its elements.

The notation $a \in A$ denotes that a is an element of the set A .

If a is not a member of A , write $a \notin A$

Describing a Set: Roster (名冊) Method

$$S = \{a, b, c, d\}$$

Order not important

$$S = \{a, b, c, d\} = \{b, c, a, d\} \quad (\text{因為「順序」不重要!})$$

Each distinct object is either a member or not; **listing more than once does not change the set.**

$$S = \{a, b, c, d\} = \{a, b, c, b, c, d\}$$

Elipses (省略符號) (...) may be used to describe a set without listing all of the members **when the pattern is clear.**

$$S = \{a, b, c, d, \dots, z\}$$

Roster Method

Set of all ^(母音)**vowels** in the English alphabet:

$$V = \{a, e, i, o, u\}$$

Set of all **odd positive integers less than 10**:

$$O = \{1, 3, 5, 7, 9\}$$

Set of all **positive integers less than 100**:

$$S = \{1, 2, 3, \dots, 99\}$$

Set of all **integers less than 0**:

$$S = \{\dots, -3, -2, -1\}$$

Some Important Sets

N = *natural numbers* = $\{0, 1, 2, 3, \dots\}$

Z = *integers* = $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

Z⁺ = *positive integers* = $\{1, 2, 3, \dots\}$

Q = set of rational numbers

R = set of *real numbers*

R⁺ = set of *positive real numbers*

C = set of *complex numbers*.

$\mathbb{N} = \{0, 1, 2, 3, \dots\}$, the set of all **natural numbers**
 $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$, the set of all **integers**
 $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$, the set of all **positive integers**
 $\mathbb{Q} = \{p/q \mid p \in \mathbb{Z}, q \in \mathbb{Z}, \text{ and } q \neq 0\}$, the set of all **rational numbers**
R, the set of all **real numbers**
R⁺, the set of all **positive real numbers**
C, the set of all **complex numbers**.

Set-Builder Notation

Specify the property or properties that all members must satisfy:

$$S = \{x \mid x \text{ is a positive integer less than } 100\}$$

$$O = \{x \mid x \text{ is an odd positive integer less than } 10\}$$

$$O = \{x \in \mathbf{Z}^+ \mid x \text{ is odd and } x < 10\}$$

A **predicate** may be used:

$$S = \{x \mid \mathbf{P}(x)\}$$

← 命題函數

Example: $S = \{x \mid \mathbf{Prime}(x)\}$

Positive rational numbers:

$$\mathbf{Q}^+ = \{x \in \mathbf{R} \mid x = p/q, \text{ for some positive integers } p, q\}$$

Interval Notation

$$[a, b] = \{x | a \leq x \leq b\}$$

$$[a, b) = \{x | a \leq x < b\}$$

$$(a, b] = \{x | a < x \leq b\}$$

$$(a, b) = \{x | a < x < b\}$$

closed interval $[a, b]$

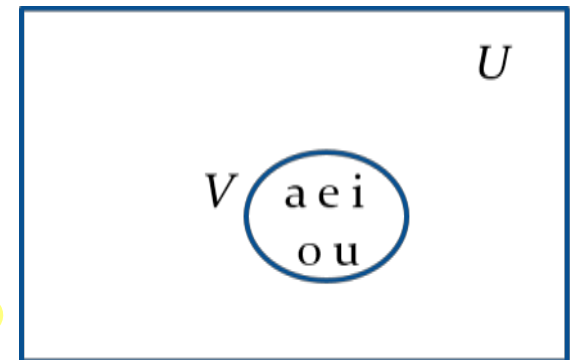
open interval (a, b)

Universal Set and Empty Set

The **universal set** (字集) U is the set containing **everything** currently under consideration.

- Sometimes implicit
- Sometimes explicitly stated.
- Contents depend on the context.

Venn Diagram (范氏圖)



The **empty set** (空集) is the set with **no elements**. Symbolized $\emptyset$, but $\{\}$ also used.

Some things to remember

Sets can be elements of sets.

$$S = \{\{1,2,3\}, a, \{b,c\}\}$$

只有这三个 ∈ S

集合内的东西
就是只能用
逗號隔开!

$\{N, Z, Q, R\}$

The empty set is different from a set containing the empty set.

$$\emptyset \neq \{\emptyset\}$$

↓
一个集合

↘ 一个集合内有一个空集合!

$$\textcircled{1} a \in S \quad \text{True}$$

$$\textcircled{2} b \in S \quad \text{False}$$

$$\textcircled{3} 1 \in S \quad \text{False}$$

$$\textcircled{4} \{1,2,3\} \in S \quad \text{True}$$

$$\textcircled{5} \{b\} \in S \quad \text{False}$$

$$\textcircled{6} \{b,c\} \in S \quad \text{True}$$

Set Equality

Definition: Two sets are *equal* if and only if they have the same elements.

- Therefore if A and B are sets, then A and B are equal if and only if $\forall x (x \in A \leftrightarrow x \in B)$ ← 證明誰是誰的子集合!
- We write $A = B$ if A and B are equal sets.

$$\{1, 3, 5\} = \{3, 5, 1\}$$

$$\{1, 5, 5, 5, 3, 3, 1\} = \{1, 3, 5\}$$

Subsets

$$\textcircled{1} \quad \emptyset \subseteq S \quad \text{True}$$

$$\textcircled{2} \quad S \subseteq S \quad \text{True}$$

Definition: The set A is a **subset** (子集合) of B , if and only if **every element of A is also an element of B .**

- The notation **$A \subseteq B$** is used to indicate that A is a subset of the set B .
- $A \subseteq B$ holds if and only if $\forall x (x \in A \rightarrow x \in B)$ is true.
 - Because $a \in \emptyset$ is always false, $\emptyset \subseteq S$, for every set S .
 - Because $a \in S \rightarrow a \in S$, $S \subseteq S$, for every set S .

13 Showing a Set is or is not a Subset of Another Set

Showing that A is a Subset of B: To show that $A \subseteq B$, show that if x belongs to A , then x also belongs to B .

Showing that A is not a Subset of B: To show that A is not a subset of B , $A \not\subseteq B$, find an element $x \in A$ with $x \notin B$. (Such an x is a counterexample to the claim that $x \in A$ implies $x \in B$.)

Examples:

1. The set of all computer science majors at your school is a subset of all students at your school.
2. The set of integers with squares less than 100 is not a subset of the set of nonnegative integers.

$$\checkmark \quad (-1)^2 < 100$$

Another look at Equality of Sets

Recall that two sets A and B are **equal**, denoted by **$A = B$** , iff

$$\forall x (x \in A \leftrightarrow x \in B)$$

Using **logical equivalences** we have that $A = B$ iff

$$\forall x [(x \in A \rightarrow x \in B) \wedge (x \in B \rightarrow x \in A)]$$

This is **equivalent to** (互相是对方的子集合)

$$A \subseteq B \quad \text{and} \quad B \subseteq A$$

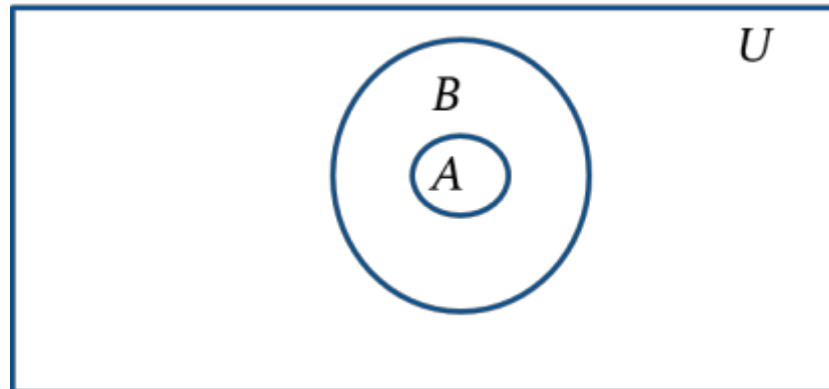
Proper Subsets

Definition: If $A \subseteq B$, but $A \neq B$, then we say A is a **proper subset** of B , denoted by $A \subset B$. If $A \subset B$, then 真的 子集合

$$\forall x(x \in A \rightarrow x \in B) \wedge \exists x(x \in B \wedge x \notin A)$$

is true.

Venn Diagram



Set Cardinality

基數

Definition: If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is **finite**. Otherwise it is **infinite**.

有限

Definition: The **cardinality** of a finite set A , denoted by $|A|$, is the number of (distinct) elements of A .

集合中的不重複元素各數

Examples:

1. $|\emptyset| = 0$

2. Let S be the letters of the English alphabet. Then $|S| = 26$

3. $|\{1, 2, 3\}| = 3$

$6 |\{\{1, 2, 3\}\}| = 1$

4. $|\{\emptyset\}| = 1$

$7 |\{\{1, 2, 3\}, \{4, 5, 6\}\}| = 2$

5. The set of integers is infinite.

Power Sets (冪集)

Definition: The set of **all subsets of a set A** , denoted $P(A)$, is called the *power set* of A .

Example: If $A = \{a, b\}$ then

$$P(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

If a set has n elements, then the cardinality of the power set is 2^n . (In Chapters 5 and 6, we will discuss different ways to show this.)

Tuples (多元組)

The **ordered n -tuple** $(a_1, a_2, \dots, a_n)$ is the ordered collection that has a_1 as its first element and a_2 as its second element and so on until a_n as its last element.

Two n -tuples are equal if and only if their corresponding elements are equal.

2-tuples are called **ordered pairs**.

The ordered pairs (a, b) and (c, d) are equal if and only if $a = c$ and $b = d$.

Cartesian Product₁

Definition: The **Cartesian Product** (笛卡兒積) of two sets A and B , denoted by $A \times B$ is the set of ordered pairs (a,b) where $a \in A$ and $b \in B$.

$$A \times B = \{(a,b) \mid a \in A \wedge b \in B\}$$

Example:

$$A = \{a,b\} \quad B = \{1,2,3\}$$

$$A \times B = \{(a,1),(a,2),(a,3), (b,1),(b,2),(b,3)\}$$

Definition: A subset R of the Cartesian product $A \times B$ is called a **relation** from the set A to the set B . (Relations will be covered in depth in Chapter 9.)

關係

Cartesian Product₂

Definition: The **Cartesian products** of the sets $A_1, A_2, \dots, A_n$, denoted by $A_1 \times A_2 \times \dots \times A_n$, is the set of ordered n -tuples $(a_1, a_2, \dots, a_n)$ where a_i belongs to A_i for $i = 1, \dots, n$.

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) \mid a_i \in A_i \text{ for } i = 1, 2, \dots, n\}$$

Example: What is $A \times B \times C$ where $A = \{0,1\}$, $B = \{1,2\}$ and $C = \{0,1,2\}$

Solution: $A \times B \times C = \{(0,1,0), (0,1,1), (0,1,2), (0,2,0), (0,2,1), (0,2,2), (1,1,0), (1,1,1), (1,1,2), (1,2,0), (1,2,1), (1,2,2)\}$

Truth Sets of Quantifiers

Given a predicate P and a domain D , we define the **truth set** of P to be the set of elements in D for which $P(x)$ is true. The truth set of $P(x)$ is denoted by

$$\{x \in D \mid P(x)\}$$

Domain

Example: The truth set of $P(x)$ where the domain is the integers and $P(x)$ is “ $|x| = 1$ ” is the set $\{-1, 1\}$

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Union

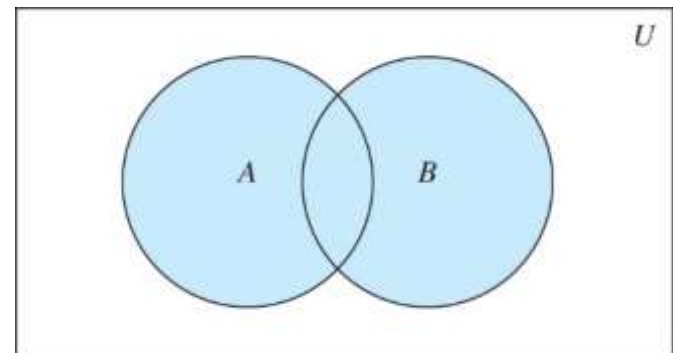
Definition: Let A and B be sets. The **union** (聯集) of the sets A and B , denoted by $A \cup B$, is the set:

$$\{x \mid x \in A \vee x \in B\}$$

Example: What is $\{1,2,3\} \cup \{3,4,5\}$?

Solution: $\{1,2,3,4,5\}$

Venn Diagram for $A \cup B$



Intersection

Definition: The **intersection** (交集) of sets A and B , denoted by $A \cap B$, is

$$\{x \mid x \in A \wedge x \in B\}$$

Note **if the intersection is empty**, then A and B are said to be **disjoint** (無交集).

Example: What is? $\{1,2,3\} \cap \{3,4,5\}$?

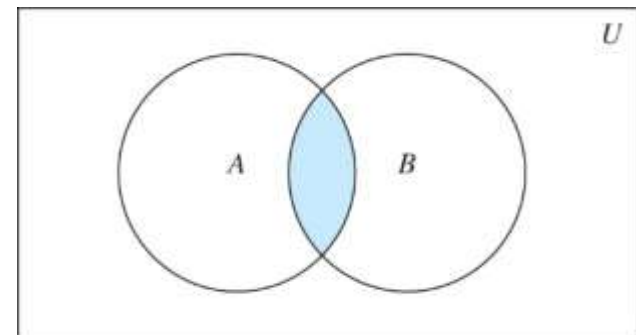
Solution: $\{3\}$

Example: What is?

$\{1,2,3\} \cap \{4,5,6\}$?

Solution: $\emptyset$

Venn Diagram for $A \cap B$



Complement (補集)

Definition: If A is a set, then the **complement** of the A (with respect to U), denoted by $\bar{A}$ is the set $U - A$

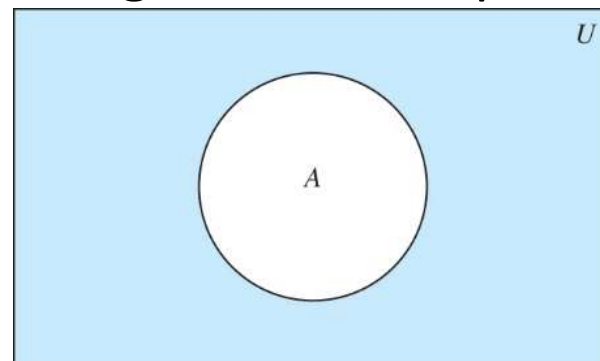
$$\bar{A} = \{x \in U \mid x \notin A\}$$

(The complement of A is sometimes denoted by A^c .)

Example: If U is the positive integers less than 100, what is the complement of $\{x \mid x > 70\}$

Solution : $\{x \mid x \leq 70\}$

Venn Diagram for Complement

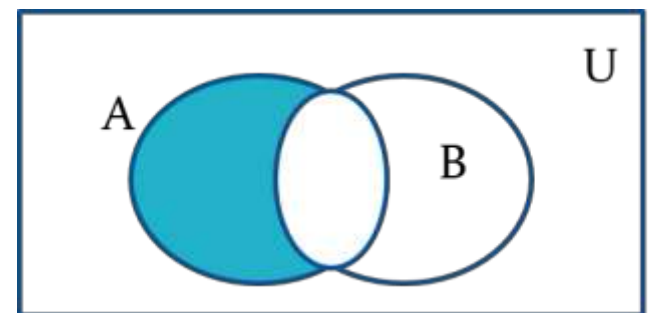


Difference (差集)

Definition: Let A and B be sets. The *difference* of A and B , denoted by $A - B$, is the set containing **the elements of A that are not in B** . The difference of A and B is also called the complement of B with respect to A .

$$A - B = \{x \mid x \in A \wedge x \notin B\} = A \cap \overline{B}$$

Venn Diagram for $A - B$



27 The Cardinality of the Union of Two Sets

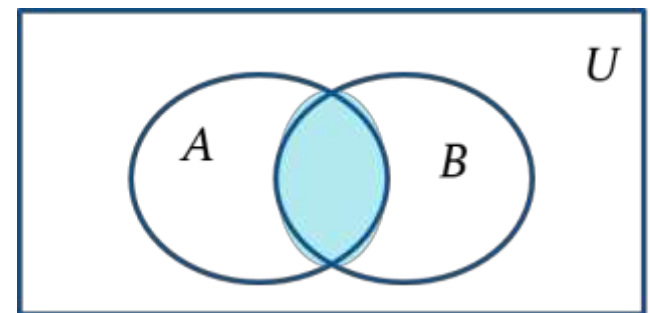
Inclusion-Exclusion

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Example: Let A be the math majors in your class and B be the CS majors. To count the number of students who are either math majors or CS majors, add the number of math majors and the number of CS majors, and subtract the number of joint CS/math majors.

We will return to this principle in Chapter 6 and Chapter 8 where we will derive a formula for the cardinality of the union of n sets, where n is a positive integer.

Venn Diagram for A , B , $A \cap B$, $A \cup B$



Review Questions

Example: $U = \{0,1,2,3,4,5,6,7,8,9,10\}$ $A = \{1,2,3,4,5\}$, $B = \{4,5,6,7,8\}$

1. $A \cup B$

Solution: $\{1,2,3,4,5,6,7,8\}$

2. $A \cap B$

Solution: $\{4,5\}$

3. $\bar{A}$

Solution: $\{0,6,7,8,9,10\}$

4. $\bar{B}$

Solution: $\{0,1,2,3,9,10\}$

5. $A - B$

Solution: $\{1,2,3\}$

6. $B - A$

Solution: $\{6,7,8\}$

Set Identities₁

Identity laws

$$A \cup \emptyset = A \quad A \cap U = A$$

Domination laws

$$A \cup U = U \quad A \cap \emptyset = \emptyset$$

Idempotent laws

$$A \cup A = A \quad A \cap A = A$$

Complementation law

$$\left(\overline{\overline{A}}\right) = A$$

Set Identities₂

Commutative laws

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

Associative laws

$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

Distributive laws

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Set Identities₃

De Morgan's laws

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

Absorption laws

$$A \cup (A \cap B) = A$$

$$A \cap (A \cup B) = A$$

Complement laws

$$A \cup \overline{A} = U$$

$$A \cap \overline{A} = \emptyset$$

Proving Set Identities

Different ways to prove set identities:

1. Prove that each set (side of the identity) is a **subset** of the other.
2. Use **set builder** notation and propositional logic.
3. Membership **Tables**: Verify that elements in the same combination of sets always either belong or do not belong to the same side of the identity. Use 1 to indicate it is in the set and a 0 to indicate that it is not

Proof of Second De Morgan Law₁

Example: Prove that $\overline{A \cap B} = \overline{A} \cup \overline{B}$

Solution: We prove this identity by showing that:

$$1) \overline{A \cap B} \subseteq \overline{A} \cup \overline{B} \quad \text{and}$$

$$2) \overline{A} \cup \overline{B} \subseteq \overline{A \cap B}$$

Proof of Second De Morgan Law₂

These steps show that: $\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$

$x \in \overline{A \cap B}$ by assumption

$x \notin A \cap B$ defn. of complement

$\neg((x \in A) \wedge (x \in B))$ by defn. of intersection

$\neg(x \in A) \vee \neg(x \in B)$ 1st De Morgan law for Prop Logic

$x \notin A \vee x \notin B$ defn. of negation

$x \in \overline{A} \vee x \in \overline{B}$ defn. of complement

$x \in \overline{A} \cup \overline{B}$ by defn. of union

Proof of Second De Morgan Law₃

These steps show that: $\overline{A \cup B} \subseteq \overline{A} \cap \overline{B}$

$$x \in \overline{A \cup B}$$

by assumption

$$(x \in \overline{A}) \vee (x \in \overline{B})$$

by defn. of union

$$(x \notin A) \vee (x \in \overline{B})$$

defn. of complement

$$\neg(x \in A) \vee \neg(x \in B)$$

defn. of negation

$$\neg((x \in A) \wedge \neg(x \in B))$$

1st De Morgan law for Prop Logic

$$\neg(x \in A \cap B)$$

defn. of intersection

$$x \in \overline{A \cap B}$$

defn. of complement

更正：沒有這個符號

Set-Builder Notation:

Second De Morgan Law

$$\begin{aligned}
 \overline{A \cap B} &= \{x | x \notin A \cap B\} && \text{by defn. of complement} \\
 &= \{x | \neg(x \in (A \cap B))\} && \text{by defn. of does not belong symbol} \\
 &= \{x | \neg(x \in A \wedge x \in B)\} && \text{by defn. of intersection} \\
 &= \{x | \neg(x \in A) \vee \neg(x \in B)\} && \text{by 1st De Morgan law for} \\
 &&& \text{Prop Logic} \\
 &= \{x | x \notin A \vee x \notin B\} && \text{by defn. of not belong symbol} \\
 &= \{x | x \in \overline{A} \vee x \in \overline{B}\} && \text{by defn. of complement} \\
 &= \{x | x \in \overline{A} \cup \overline{B}\} && \text{by defn. of union} \\
 &= \overline{A} \cup \overline{B} && \text{by meaning of notation}
 \end{aligned}$$

Membership Table

暴力法

Example: Construct a membership table to show that the distributive law holds.

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Solution:

A	B	C	$B \cap C$	$A \cup (B \cap C)$	$A \cup B$	$A \cup C$	$(A \cup B) \cap (A \cup C)$
1	1	1	1	1	1	1	1
1	1	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	0	0	0	1	1	1	1
0	1	1	1	1	1	1	1
0	1	0	0	0	1	0	0
0	0	1	0	0	0	1	0
0	0	0	0	0	0	0	0

Generalized Unions and Intersections

Let $A_1, A_2, \dots, A_n$ be an indexed collection of sets.

We define:

$$\bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup \dots \cup A_n$$

$$\bigcap_{i=1}^n A_i = A_1 \cap A_2 \cap \dots \cap A_n$$

These are well defined, since union and intersection are associative.

For $i = 1, 2, \dots$, let $A_i = \{i, i + 1, i + 2, \dots\}$. Then,

$$\bigcup_{i=1}^n A_i = \bigcup_{i=1}^n \{i, i + 1, i + 2, \dots\} = \{1, 2, 3, \dots\}$$

$$\bigcap_{i=1}^n A_i = \bigcap_{i=1}^n \{i, i + 1, i + 2, \dots\} = \{n, n + 1, n + 2, \dots\} = A_n$$

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Functions₁

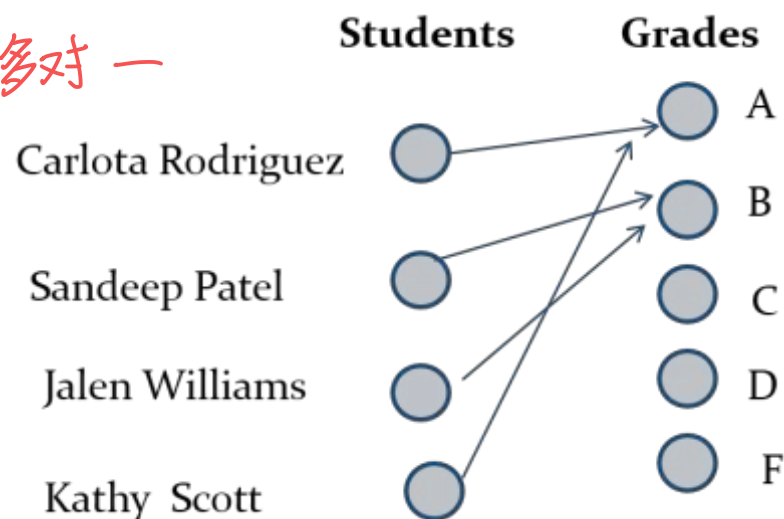
從定義出發！

Definition: Let A and B be nonempty sets. A **function** f from A to B , denoted $f: A \rightarrow B$ is an assignment of each element of A to exactly one element of B . We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A .

- **Functions** are sometimes called **mappings** or **transformations**.

一對一 or

多對一

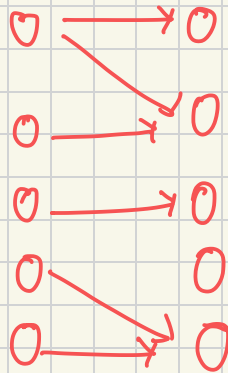


Review: Functions or NoT?

<key: 一对一 or 多对一>

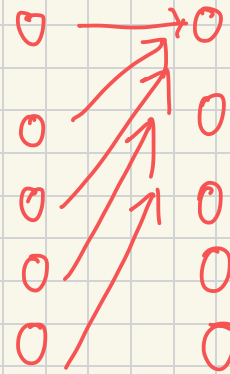
(a)

X



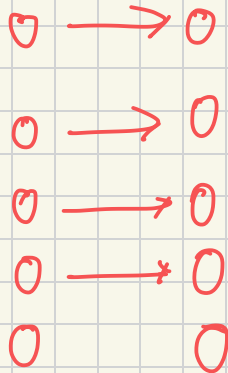
(b)

✓



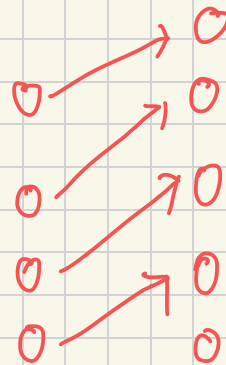
(c)

X



(d)

✓



Functions₂

$$\begin{array}{l} (2, 9) \\ f(2) = 9 \end{array} \quad f: 2 \rightarrow 9$$

A function $f: A \rightarrow B$ can also be defined as a subset of $A \times B$ (a relation). This subset is restricted to be a relation where no two elements of the relation have the same first element.

Specifically, a function f from A to B contains one, and only one ordered pair (a, b) for every element $a \in A$.

$$\forall x \left[x \in A \rightarrow \exists y \left[y \in B \wedge (x, y) \in f \right] \right]$$

and

$$\forall x, y_1, y_2 \left[\left[(x, y_1) \in f \wedge (x, y_2) \in f \right] \rightarrow y_1 = y_2 \right]$$

Functions₃

Given a function $f: A \rightarrow B$:

- We say f maps A to B or f is a *mapping* from A to B .

- A is called the **domain** of f .

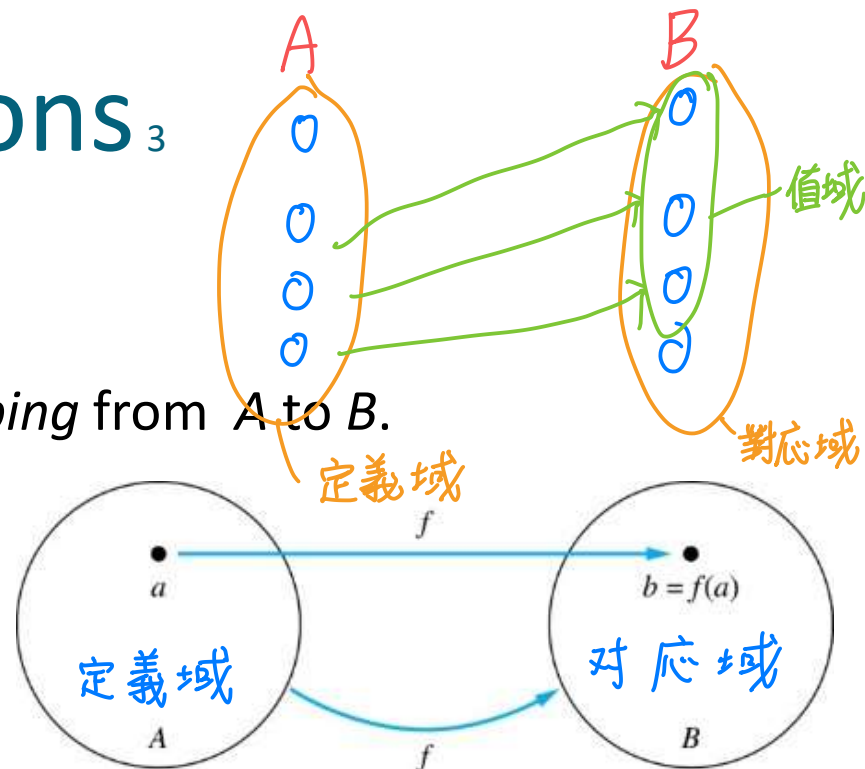
- B is called the **codomain** of f .

- If $f(a) = b$,
 - then b is called the **image** of a under f .

- a is called the **preimage** of b .

- The range of f is the set of all images of points in A under f . We denote it by $f(A)$.

- Two functions are **equal** when they have the same domain, the same codomain and map each element of the domain to the same element of the codomain.



Representing Functions

Functions may be specified in different ways:

- An explicit statement of the assignment. Students and grades example.

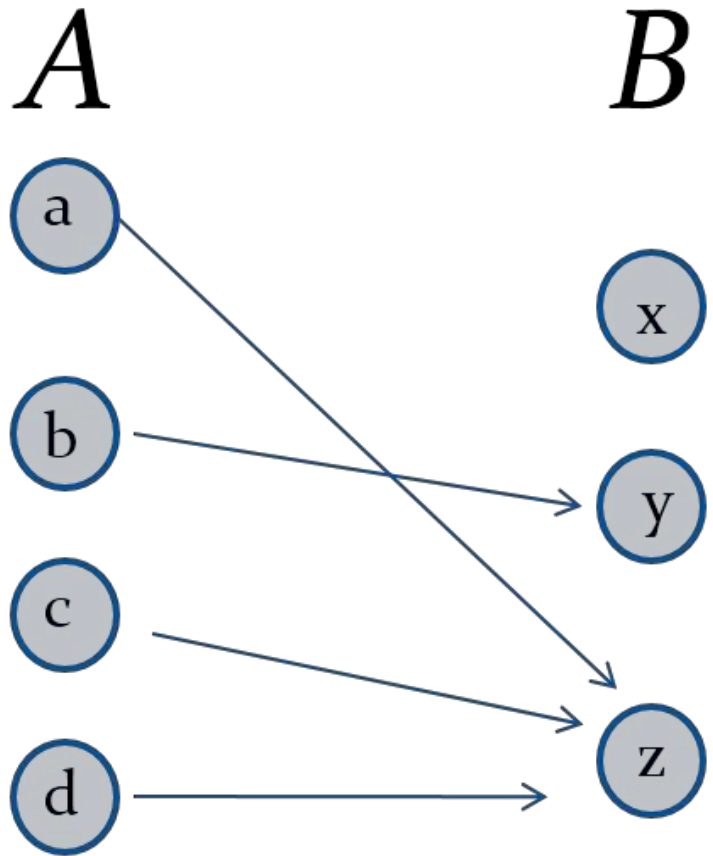
- A formula.

$$f(x) = x + 1$$

- A computer program.

- A Java program that when given an integer n , produces the n th Fibonacci Number (covered in the next section and also in Chapter 5).

Questions



$f(a) = ?$ z

The image of d is ? z

The domain of f is ? A

The codomain of f is ? B

The preimage of y is ? b

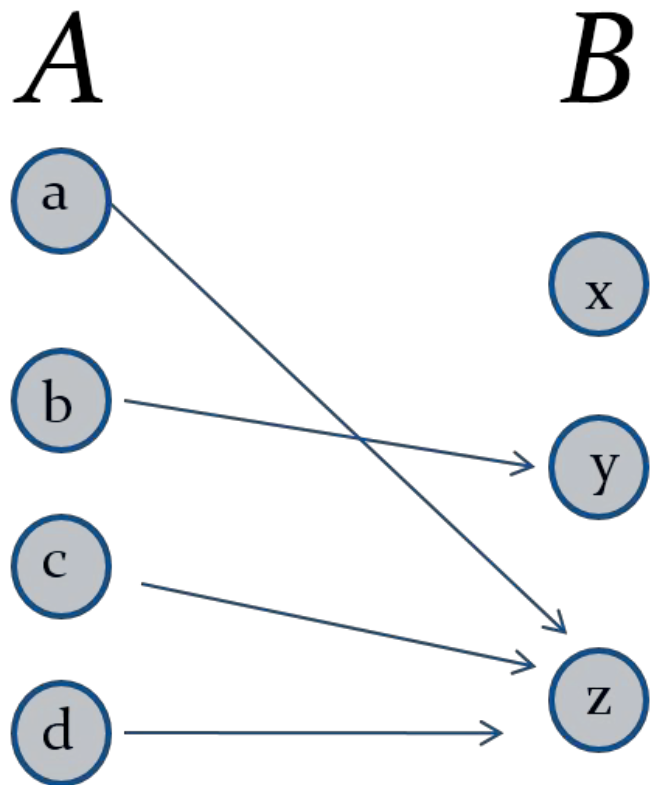
$f(A) = ?$ $\{y, z\}$

The preimage(s) of z is (are) ? $\{a, c, d\}$

Question on Functions and Sets

If $f: A \rightarrow B$ and S is a subset of A , then

$$f(S) = \{f(s) \mid s \in S\}$$



$f(\{a, b, c\})$ is ? $\{y, z\}$

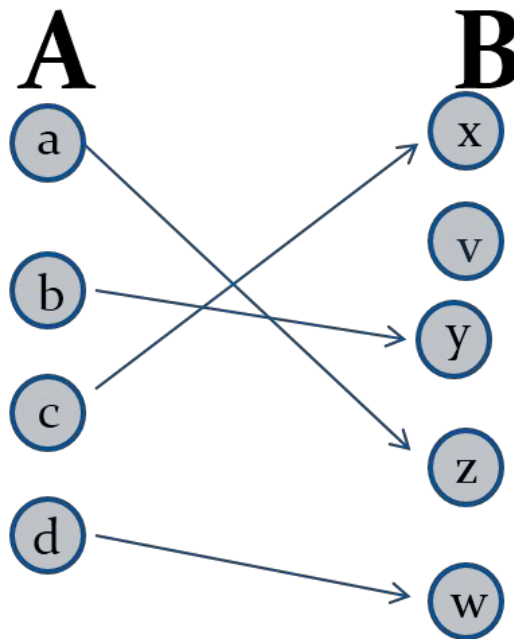
$f(\{c, d\})$ is ? $\{z\}$

Injections

又寫作 1-1

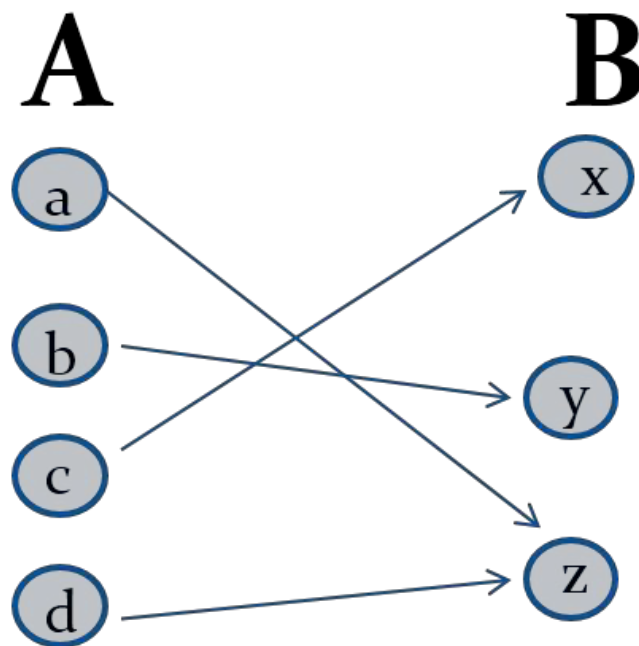
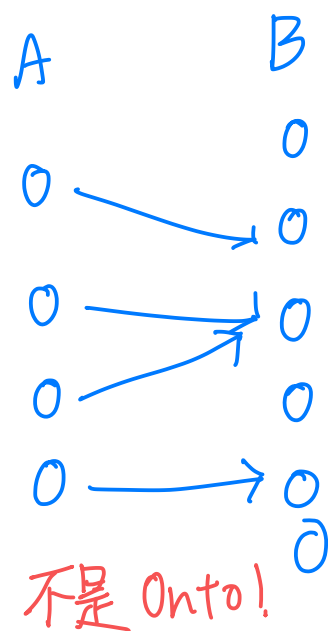
Definition: A function f is said to be **one-to-one**, or **injective**, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f . A function is said to be an *injection* if it is one-to-one.

就是
一對一！
不能有多對一的 Case！



Surjections

Definition: A function f from A to B is called **onto** or **surjective**, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$.
A function f is called a *surjection* if it is *onto*.



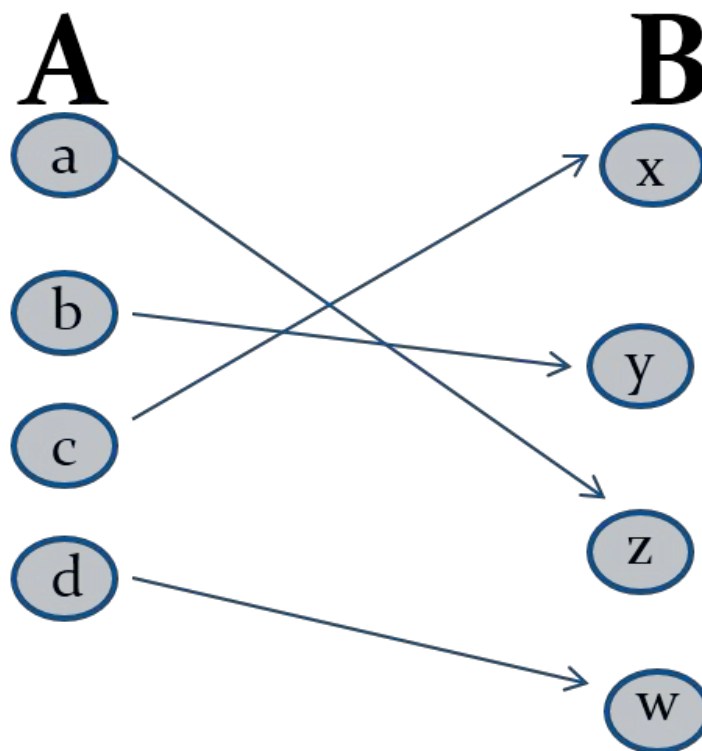
就是
对应域
全覆盖!

Bijections

(双向互射)

Definition: A function f is a one-to-one correspondence, or a bijection, if it is both one-to-one and onto (surjective and injective).

一对一 且
全覆盖



A B

0 → 0

0 → 0

0 → 0

0 → 0

0 → 0

X 不是 1-1 and onto

Showing that f is one-to-one or onto₁

Suppose that $f: A \rightarrow B$.

To show that f is injective ⁽¹⁻¹⁾ Show that if $f(x) = f(y)$ for arbitrary $x, y \in A$, then $x = y$.

To show that f is not injective ^(不是 1-1) Find particular elements $x, y \in A$ such that $x \neq y$ and $f(x) = f(y)$.

To show that f is surjective ^(是 onto) Consider an arbitrary element $y \in B$ and find an element $x \in A$ such that $f(x) = y$.

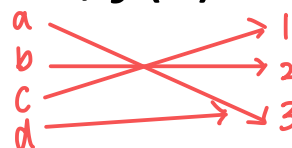
To show that f is not surjective ^(不是 onto) Find a particular $y \in B$ such that $f(x) \neq y$ for all $x \in A$.

Showing that f is one-to-one or onto₂

Example 1: Let f be the function from $\{a,b,c,d\}$ to $\{1,2,3\}$ defined by $f(a) = 3$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$.

Is f an onto function?

是 onto!



Solution: Yes, f is onto since **all three elements** of the codomain are images of elements in the domain. If the codomain were changed to $\{1,2,3,4\}$, f would not be onto.

Example 2: Is the function $f(x) = x^2$ from the set of integers to the set of integers onto?

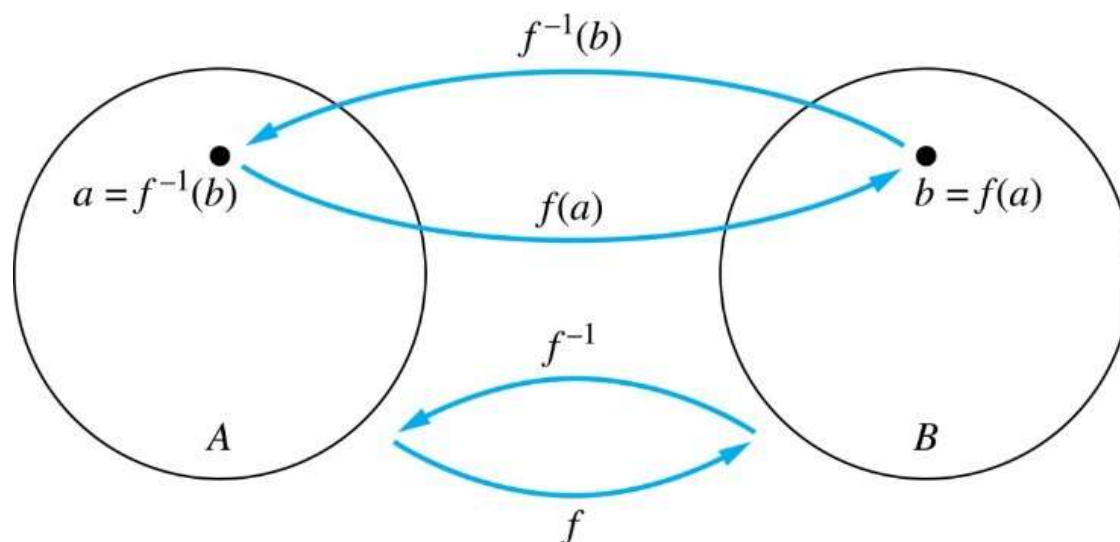
Solution: No, f is not onto because **there is no integer x with $x^2 = -1$** , for example.

Inverse Functions₁

↙ 先证是 1-1 and onto

Definition: Let f be a **bijection** from A to B . Then the **inverse** of f , denoted f^{-1} , is the function from B to A defined as $f^{-1}(y) = x$ iff $f(x) = y$

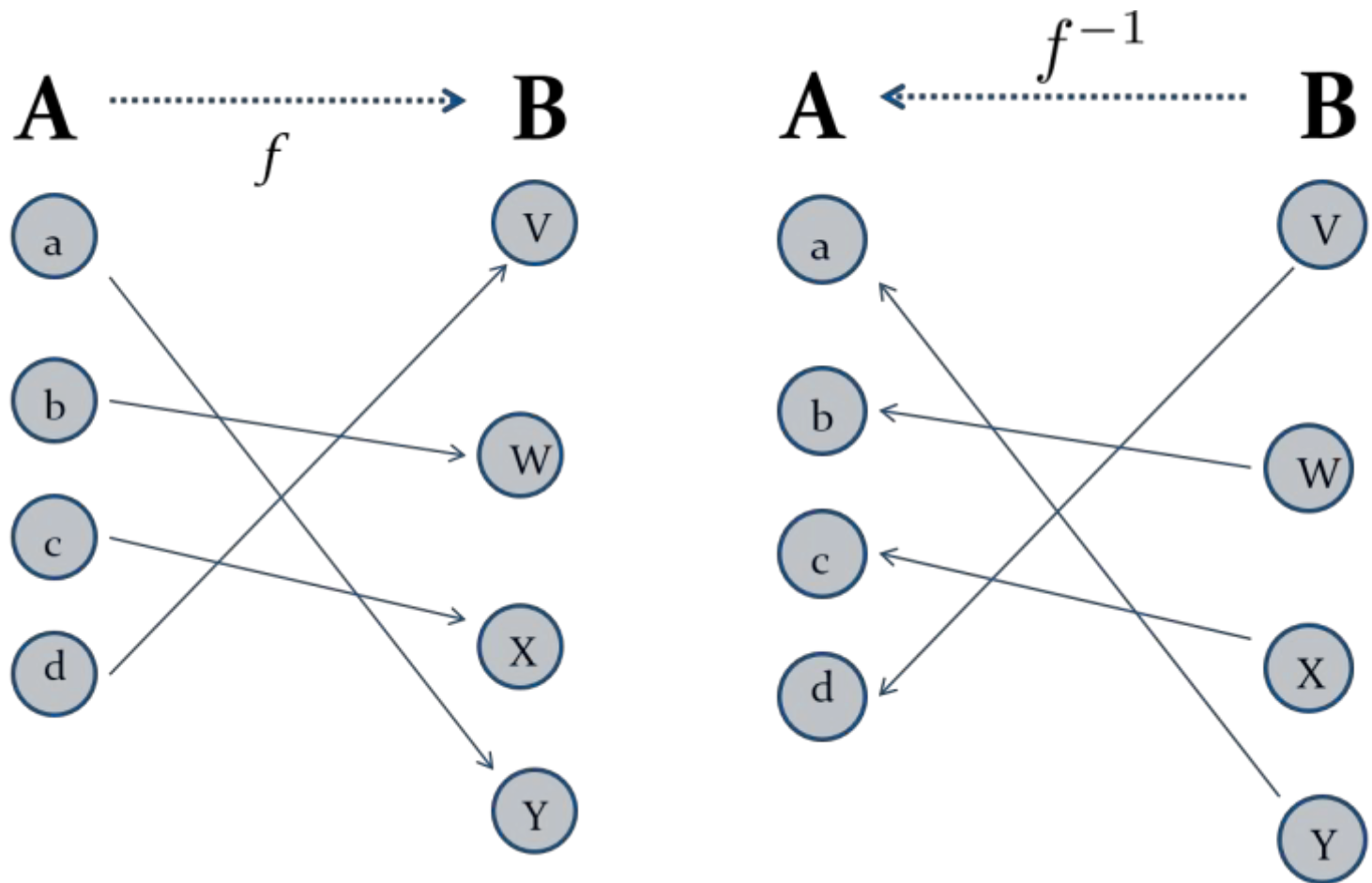
No inverse exists unless f is a bijection. Why?



[Jump to long description](#)

Inverse Functions₂

反函數



Questions₁

Example 1: Let f be the function from $\{a,b,c\}$ to $\{1,2,3\}$ such that $f(a) = 2$, $f(b) = 3$, and $f(c) = 1$.

Is f invertible and if so what is its inverse?

Solution: The function f is invertible because it is a one-to-one correspondence. The inverse function f^{-1} reverses the correspondence given by f , so $f^{-1}(1) = c$, $f^{-1}(2) = a$, and $f^{-1}(3) = b$.

Questions₂

Example 2: Let $f: \mathbf{Z} \rightarrow \mathbf{Z}$ be such that $f(x) = x + 1$.
Is f invertible, and if so, what is its inverse?

$$\begin{aligned} \text{let } f(x) &= y \\ f(x) &= x + 1 \\ y &= x + 1 \\ x &= y - 1 \end{aligned}$$

Solution: The function f is invertible because it is a one-to-one correspondence. The inverse function f^{-1} reverses the correspondence so $f^{-1}(y) = y - 1$.

Questions₃

1, -1 結果都是 1)
 ∴ 不是 1-1 and onto!

Example 3: Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be such that $f(x) = x^2$

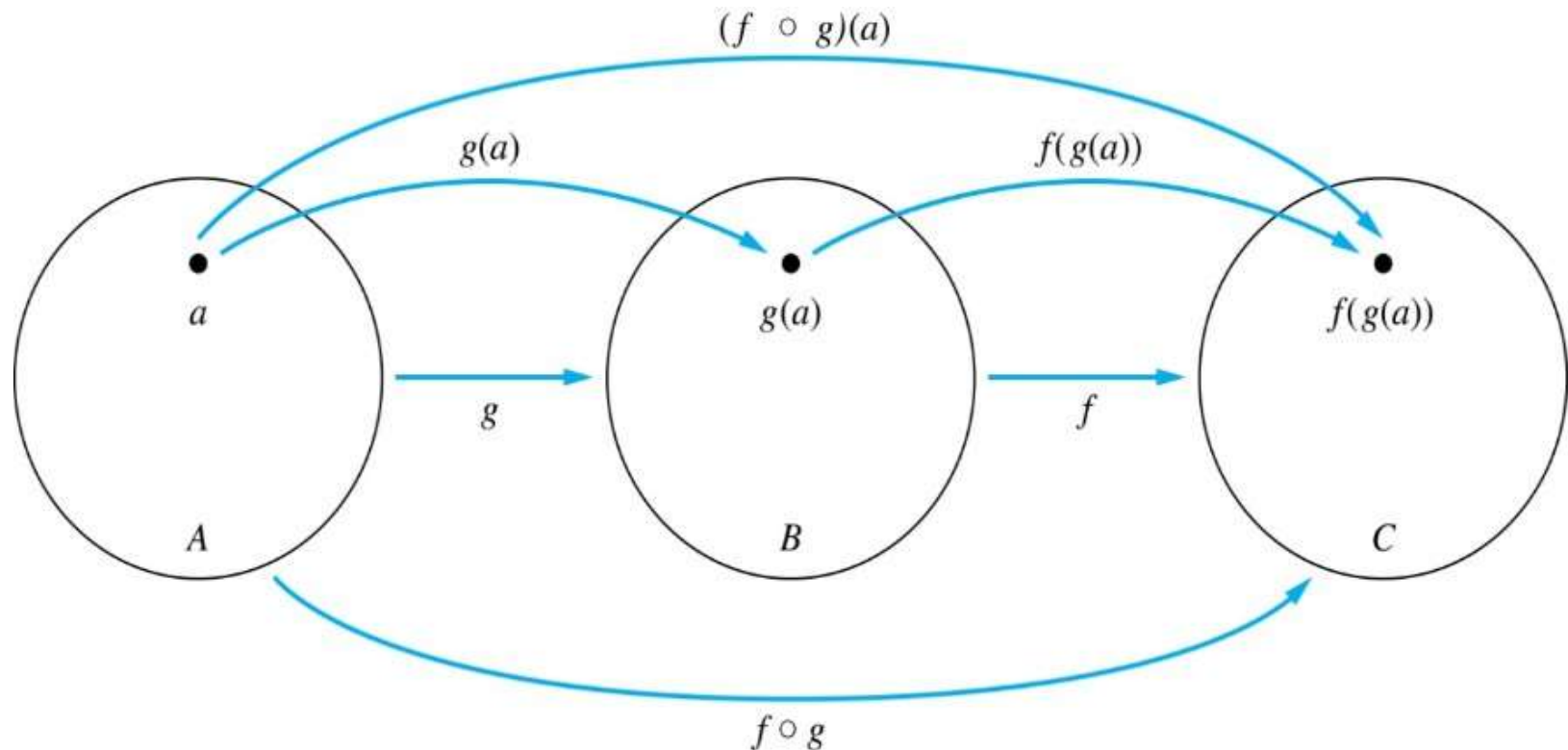
Is f invertible, and if so, what is its inverse?

Solution: The function f is not invertible **because it is not one-to-one.**

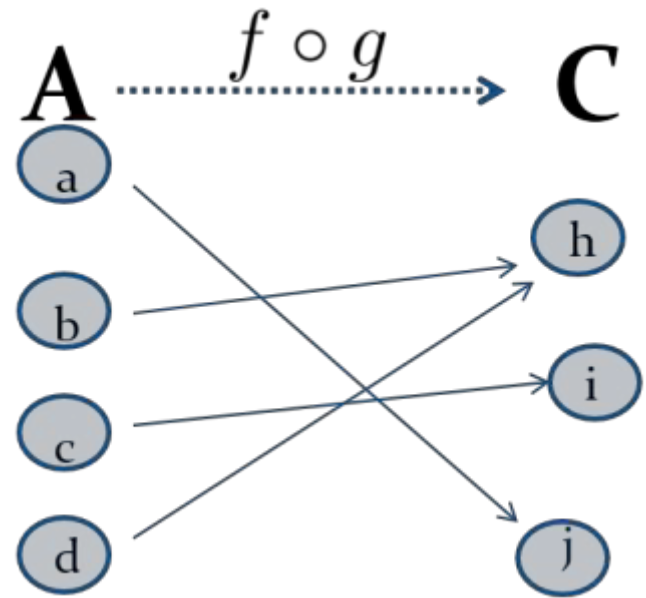
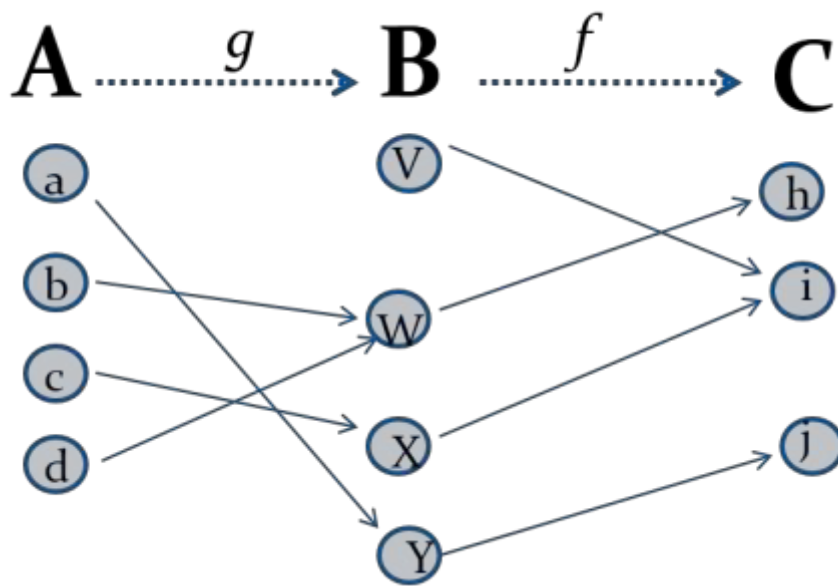
Composition₁(合成)

Definition: Let $f: B \rightarrow C, g: A \rightarrow B$. The **composition** of f with g , denoted $f \circ g$ is the function from A to C defined by

$$f \circ g(x) = f(g(x))$$



Composition₂



Composition₃

Example 1: If

$$f(x) = x^2 \text{ and } g(x) = 2x + 1,$$

then

$$f(g(x)) = (2x + 1)^2$$

and

$$g(f(x)) = 2x^2 + 1$$

Composition Questions₁

Example 2: Let g be the function from the set $\{a,b,c\}$ to itself such that $g(a) = b$, $g(b) = c$, and $g(c) = a$. Let f be the function from the set $\{a,b,c\}$ to the set $\{1,2,3\}$ such that $f(a) = 3$, $f(b) = 2$, and $f(c) = 1$.

What is the composition of f and g , and what is the composition of g and f ?
 $\rightarrow f \circ g \rightarrow f(g(x))$
 $g \circ f \rightarrow g(f(x))$

Solution: The composition $f \circ g$ is defined by

$$f \circ g(a) = f(g(a)) = f(b) = 2$$

$$f \circ g(b) = f(g(b)) = f(c) = 1$$

$$f \circ g(c) = f(g(c)) = f(a) = 3$$

Note that $g \circ f$ is not defined, because the range of f is not a subset of the domain of g .

$$g \circ f(a) = g(f(a)) = g(3) = ???$$

Composition Questions₂

Example 2: Let f and g be functions from the set of integers to the set of integers defined by

$$f(x) = 2x + 3 \quad \text{and} \quad g(x) = 3x + 2.$$

What is the composition of f and g , and also the composition of g and f ?

Solution:

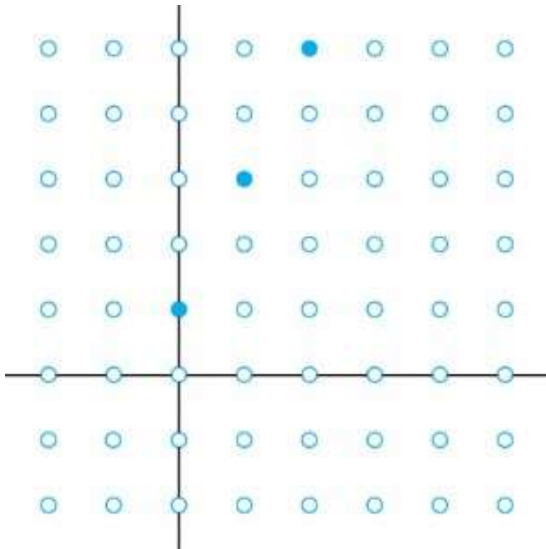
$$f \circ g(x) = f(g(x)) = f(3x + 2) = 2(3x + 2) + 3 = 6x + 7$$

$$g \circ f(x) = g(f(x)) = g(2x + 3) = 3(2x + 3) + 2 = 6x + 11$$

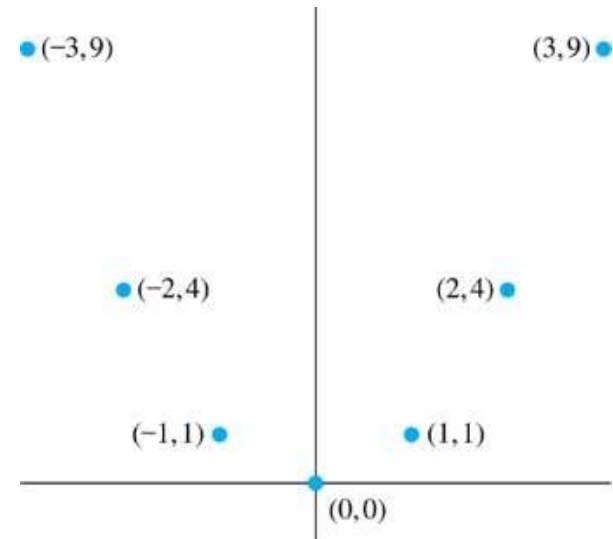
Graphs of Functions

Let f be a function from the set A to the set B . The *graph* of the function f is the set of ordered pairs

$$\{(a, b) \mid a \in A \text{ and } f(a) = b\}.$$



Graph of $f(n) = 2n + 1$
from $\mathbb{Z}$ to $\mathbb{Z}$



Graph of $f(x) = x^2$
from $\mathbb{Z}$ to $\mathbb{Z}$

[Jump to long description](#)

Some Important Functions

The **floor** function, denoted

$$f(x) = \lfloor x \rfloor$$

is the largest integer less than or equal to x .

The **ceiling** function, denoted

$$f(x) = \lceil x \rceil$$

is the smallest integer greater than or equal to x

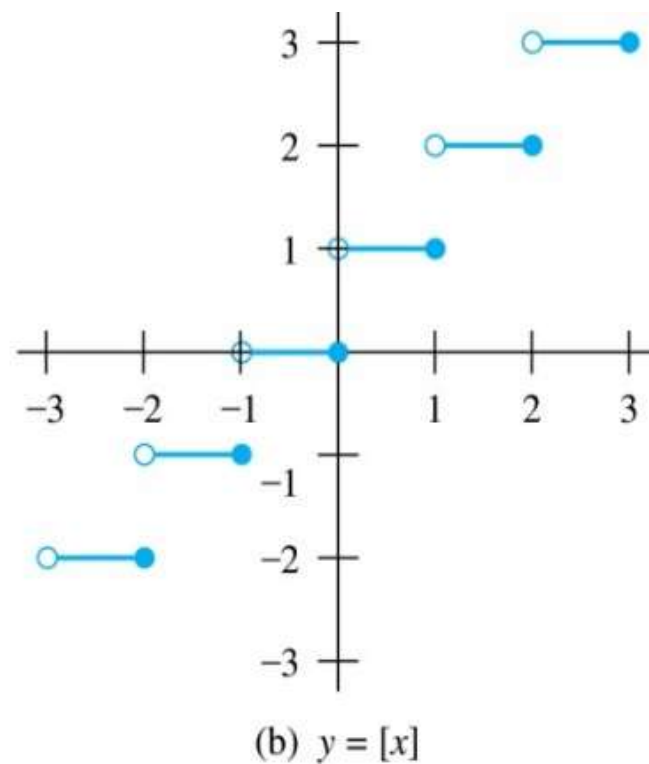
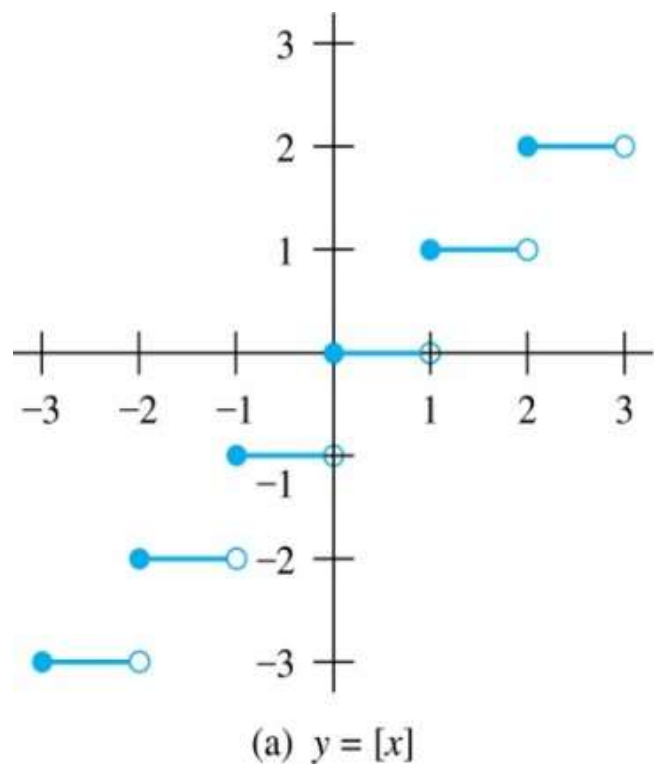
Example:

$\lceil 3.5 \rceil = 4$	$\lfloor 3.5 \rfloor = 3$
$\lceil -1.5 \rceil = -1$	$\lfloor -1.5 \rfloor = -2$

上高斯

下高斯

Floor and Ceiling Functions₁



Graph of (a) Floor and (b) Ceiling Functions

[Jump to long description](#)

Floor and Ceiling Functions₂

TABLE 1 Useful Properties of the Floor and Ceiling Functions.

(n is an integer, x is a real number)

$$(1a) \quad \lfloor x \rfloor = n \text{ if and only if } n \leq x < n + 1$$

$$(1b) \quad \lceil x \rceil = n \text{ if and only if } n - 1 < x \leq n$$

$$(1c) \quad \lfloor x \rfloor = n \text{ if and only if } x - 1 < n \leq x$$

$$(1d) \quad \lceil x \rceil = n \text{ if and only if } x \leq n < x + 1$$

$$(2) \quad x - 1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x + 1$$

$$(3a) \quad \lfloor -x \rfloor = -\lceil x \rceil$$

$$(3b) \quad \lceil -x \rceil = -\lfloor x \rfloor$$

$$(4a) \quad \lfloor x + n \rfloor = \lfloor x \rfloor + n$$

$$(4b) \quad \lceil x + n \rceil = \lceil x \rceil + n$$

Proving Properties of Functions

Example: Prove that x is a real number, then

$$\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + 1/2 \rfloor$$

Solution: Let $x = n + \varepsilon$, where n is an integer and $0 \leq \varepsilon < 1$.

Case 1: $\varepsilon < 1/2$

- $2x = 2n + 2\varepsilon$ and $\lfloor 2x \rfloor = 2n$, since $0 \leq 2\varepsilon < 1$.
- $\lfloor x + 1/2 \rfloor = n$, since $x + 1/2 = n + (1/2 + \varepsilon)$ and $0 \leq 1/2 + \varepsilon < 1$.
- Hence, $\lfloor 2x \rfloor = 2n$ and $\lfloor x \rfloor + \lfloor x + 1/2 \rfloor = n + n = 2n$.

Case 2: $\varepsilon \geq 1/2$

- $2x = 2n + 2\varepsilon = (2n + 1) + (2\varepsilon - 1)$ and $\lfloor 2x \rfloor = 2n + 1$, since $0 \leq 2\varepsilon - 1 < 1$.
- $\lfloor x + 1/2 \rfloor = \lfloor n + (1/2 + \varepsilon) \rfloor = \lfloor n + 1 + (\varepsilon - 1/2) \rfloor = n + 1$ since $0 \leq \varepsilon - 1/2 < 1$.
- Hence, $\lfloor 2x \rfloor = 2n + 1$ and $\lfloor x \rfloor + \lfloor x + 1/2 \rfloor = n + (n + 1) = 2n + 1$.

分
 $\begin{cases} > 1/2 \\ < 1/2 \end{cases}$
 兩Case

Factorial Function

階乘

Definition: $f: \mathbf{N} \rightarrow \mathbf{Z}^+$, denoted by $f(n) = n!$ is the product of the first n positive integers when n is a nonnegative integer.

$$f(n) = 1 \cdot 2 \cdots (n - 1) \cdot n$$

$$f(0) = 0! = 1$$

Examples:

$$f(1) = 1! = 1$$

$$f(2) = 2! = 1 \cdot 2$$

$$f(6) = 6! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 = 720$$

$$f(20) = 2,432,902,008,176,640,000$$

Chapter 2.

Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

2.1 Sets

2.2 Set Operations

2.3 Functions

2.4 Sequences and Summations

2.5 Cardinality of Sets

序列 & 總和

2.6 Matrices

Introduction₂

Sequences (序列) are ordered lists of elements.

- 1, 2, 3, 5, 8
- 1, 3, 9, 27, 81,

Sequences arise throughout mathematics, computer science, and in many other disciplines, ranging from botany to music.

We will introduce the terminology to represent sequences and sums of the terms in the sequences.

Sequences₁

Definition: A **sequence** is a function from a subset of the integers (usually either the set $\{0, 1, 2, 3, 4, \dots\}$ or $\{1, 2, 3, 4, \dots\}$) to a set S .

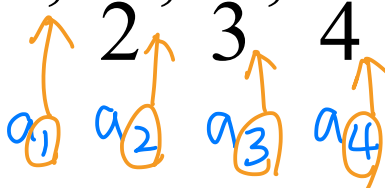
The notation a_n is used to denote the image of the integer n . We can think of a_n as the **equivalent of $f(n)$** where f is a function from $\{0, 1, 2, \dots\}$ to S . We call a_n a **term** (項) of the sequence.

Sequences₂

Example: Consider the sequence $\{a_n\}$ where

$$a_n = \frac{1}{n}$$

$$\{a_n\} = \{a_1, a_2, a_3, \dots\}$$

$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$

 $a_1, a_2, a_3, a_4, \dots$

or

$a_0, a_1, a_2, a_3, \dots$

Geometric Progression

Definition: A *geometric progression* (幾何數列) is a sequence of the form: $a, ar^2, \dots, ar^n, \dots$

where the *initial term* a and the *common ratio* r are real numbers.

Examples :

1. Let $a = 1$ and $r = -1$. Then :

$$\{b_n\} = \{b_0, b_1, b_2, b_3, b_4, \dots\} = \{1, -1, 1, -1, 1, \dots\}$$

2. Let $a = 2$ and $r = 5$. Then :

$$\{c_n\} = \{c_0, c_1, c_2, c_3, c_4, \dots\} = \{2, 10, 50, 250, 1250, \dots\}$$

3. Let $a = 6$ and $r = 1/3$. Then :

$$\{d_n\} = \{d_0, d_1, d_2, d_3, d_4, \dots\} = \left\{6, 2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \dots\right\}$$

Arithmetic Progression

Definition: A **arithmetic progression (等差數列)** is a sequence of the form: $a, a + d, a + 2d, \dots, a + nd, \dots$ where the *initial term* a and the *common difference* d are real numbers.

Examples :

1. Let $a = -1$ and $d = 4$:

$$\{s_n\} = \{s_0, s_1, s_2, s_3, s_4, \dots\} = \{-1, 3, 7, 11, \dots\}$$

2. Let $a = 7$ and $d = -3$:

$$\{t_n\} = \{t_0, t_1, t_2, t_3, t_4, \dots\} = \{7, 4, 1, -2, -5, \dots\}$$

3. Let $a = 1$ and $d = 2$:

$$\{u_n\} = \{u_0, u_1, u_2, u_3, u_4, \dots\} = \{1, 3, 5, 7, 9, \dots\}$$

Strings₁

字串(?)

Definition: A *string* is a finite sequence of characters from a finite set (an alphabet).

Sequences of characters or bits are important in computer science.

The *empty string* is represented by λ .

The string *abcde* has length 5.

Recurrence Relations

Definition: A **recurrence relation** (遞迴關係) for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms, namely, $a_0, a_1, \dots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a nonnegative integer.

A sequence is called a **solution** of a recurrence relation if its terms satisfy the recurrence relation.

The **initial conditions** (初始條件) for a sequence specify the terms that precede the first term where the recurrence relation takes effect.

Questions about Recurrence Relations₁

Example 1: Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 1, 2, 3, 4, \dots$ and suppose that $a_0 = 2$. What are a_1 , a_2 and a_3 ?

[Here $a_0 = 2$ is the initial condition.]

Solution: We see from the recurrence relation that

$$\begin{aligned}a_1 &= a_0 + 3 = 2 + 3 = 5 \\a_2 &= a_1 + 3 = 5 + 3 = 8 \\a_3 &= a_2 + 3 = 8 + 3 = 11\end{aligned}$$

Questions about Recurrence Relations₂

Example 2: Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} - a_{n-2}$ for $n = 2, 3, 4, \dots$ and suppose that $a_0 = 3$ and $a_1 = 5$. What are a_2 and a_3 ?

[Here the initial conditions are $a_0 = 3$ and $a_1 = 5$.]

Solution: We see from the recurrence relation that

$$a_2 = a_1 - a_0 = 5 - 3 = 2$$

$$a_3 = a_2 - a_1 = 2 - 5 = -3$$

Fibonacci Sequence

Definition: Define the *Fibonacci sequence*, $f_0, f_1, f_2, \dots$, by:

費波那契數列

- Initial Conditions: $f_0 = 0, f_1 = 1$
- Recurrence Relation: $f_n = f_{n-1} + f_{n-2}$

$f_n = ?$

Example: Find f_2, f_3, f_4, f_5 and f_6 .

1. 直接猜 → 證明他
2. 計算他 → 求公式

$$f_2 = f_1 + f_0 = 1 + 0 = 1$$

$$f_3 = f_2 + f_1 = 1 + 1 = 2$$

$$f_4 = f_3 + f_2 = 2 + 1 = 3$$

$$f_5 = f_4 + f_3 = 3 + 2 = 5$$

$$f_6 = f_5 + f_4 = 5 + 3 = 8$$

Solving Recurrence Relations

Finding a formula for the n th term of the sequence generated by a recurrence relation is called *solving the recurrence relation*.

Such a formula is called a *closed formula*.

Various methods for solving recurrence relations will be covered in Chapter 8 where recurrence relations will be studied in greater depth.

Here we illustrate by example the method of iteration in which we need to guess the formula. The guess can be proved correct by the method of induction (Chapter 5).

Iterative Solution Example₁

Method 1: Working upward, forward substitution Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 2, 3, 4, \dots$ and suppose that $a_1 = 2$.

從 $\rightarrow a_2 = 2 + 3$

前面 $a_3 = (2 + 3) + 3 = 2 + 3 \cdot 2$

開始寫 $a_4 = (2 + 2 \cdot 3) + 3 = 2 + 3 \cdot 3$

只要能夠猜出來，中間過程不管如何

基本上都可以接受！

$$a_n = a_{n-1} + 3 = (2 + 3 \cdot (n - 2)) + 3 = 2 + 3(n - 1)$$

Iterative Solution Example₂

Method 2: Working downward, backward substitution Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 2, 3, 4, \dots$ and suppose that $a_1 = 2$.

從後面(an) 开始寫!

$$\begin{aligned} \rightarrow a_n &= a_{n-1} + 3 \\ &= (a_{n-2} + 3) + 3 = a_{n-2} + 3 \cdot 2 \\ &= (a_{n-3} + 3) + 3 \cdot 2 = a_{n-3} + 3 \cdot 3 \end{aligned}$$

.

.

.

$$= a_2 + 3(n - 2) = (a_1 + 3) + 3(n - 2) = 2 + 3(n - 1)$$

Financial Application₁

Example: Suppose that a person deposits \$10,000.00 in a savings account at a bank yielding 11% per year with interest compounded annually. How much will be in the account after n years?

Let P_n denote the amount in the account after n years. P_n satisfies the following recurrence relation:

$$P_n = P_{n-1} + 0.11P_{n-1} = (1.11) P_{n-1}$$

with the initial condition $P_0 = 10,000$

Financial Application₂

$$P_n = P_{n-1} + 0.11P_{n-1} = (1.11) P_{n-1}$$

with the initial condition $P_0 = 10,000$

Solution: Forward Substitution

$$P_1 = (1.11)P_0$$

$$P_2 = (1.11)P_1 = (1.11)^2 P_0$$

$$P_3 = (1.11)P_2 = (1.11)^3 P_0$$

:

$$P_n = (1.11)P_{n-1} = (1.11)^n P_0 = (1.11)^n 10,000$$

$P_n = (1.11)^n 10,000$ (Can prove by induction, covered in Chapter 5)

$$P_{30} = (1.11)^{30} 10,000 = \$228,992.97$$

Summations₁

Sum of the terms $a_m, a_{m+1}, \dots, a_n$

from the sequence $\{a_n\}$

The notation:

$$\sum_{j=m}^n a_j \quad \sum_{j=m}^n a_j \quad \sum_{m \leq j \leq n} a_j$$

represents

$$a_m + a_{m+1} + \dots + a_n$$

The variable j is called the *index of summation*. It runs through all the integers starting with its *lower limit* m and ending with its *upper limit* n .

Summations₂

More generally for a set S :

$$\sum_{j \in S} a_j$$

Examples:

$$r^0 + r^1 + r^2 + r^3 + \dots + r^n = \sum_{j=0}^n r^j$$

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots = \sum_{i=1}^{\infty} \frac{1}{i}$$

If $S = \{2, 5, 7, 10\}$ then $\sum_{j \in S} a_j = a_2 + a_5 + a_7 + a_{10}$

Product Notation

Product of the terms $a_m, a_{m+1}, \dots, a_n$
 from the sequence $\{a_n\}$

The notation:

$$\prod_{j=m}^n a_j$$

$$\prod_{j=m}^n a_j$$

$$\prod_{m \leq j \leq n} a_j$$

represents

$$a_m \times a_{m+1} \times \dots \times a_n$$

Geometric Series₁

Sums of terms of geometric progressions

$$\sum_{j=0}^n ar^j = \begin{cases} \frac{ar^{n+1} - a}{r - 1} & r \neq 1 \\ (n+1)a & r = 1 \end{cases}$$

Proof: Let

$$S_n = \sum_{j=0}^n ar^j$$

$$rS_n = r \sum_{j=0}^n ar^j$$

$$= \sum_{j=0}^n ar^{j+1}$$

To compute S_n , first multiply both sides of the equality by r and then manipulate the resulting sum as follows:

Geometric Series₂

$$= \sum_{j=0}^n ar^{j+1}$$

From previous slide.

$$= \sum_{k=1}^{n+1} ar^k$$

Shifting the index of summation with $k = j + 1$.

$$= \left(\sum_{k=0}^n ar^k \right) + (ar^{n+1} - a)$$

Removing $k = n + 1$ term and adding $k = 0$ term.

$$= S_n + (ar^{n+1} - a)$$

Substituting S for summation formula

$$\therefore rS_n = S_n + (ar^{n+1} - a)$$

$$S_n = \frac{ar^{n+1} - a}{r - 1}$$

if $r \neq 1$

$$S_n = \sum_{j=0}^n ar^j = \sum_{j=0}^n a = (n+1)a$$

if $r = 1$

Some Useful Summation Formulae

常用級數公式!

TABLE 2 Some Useful Summation Formulae.

Sum	Closed Form
$\sum_{k=0}^n ar^k \ (r \neq 0)$	$\frac{ar^{n+1} - a}{r - 1}, \ r \neq 1$
$\sum_{k=1}^n k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^n k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^n k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k, x < 1$	$\frac{1}{1-x}$
$\sum_{k=0}^{\infty} kx^{k-1}, x < 1$	$\frac{1}{(1-x)^2}$

Geometric Series:
We have proved this.

Later we will prove
some of these by
induction.

Proof in text
(requires calculus)

Chapter 2.

Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

2.1 Sets

本章

2.2 Set Operations

超重要!

2.3 Functions

2.4 Sequences and Summations

2.5 Cardinality of Sets

2.6 Matrices 基數

Cardinality₁

Definition: The **cardinality** of a set A is equal to the cardinality of a set B , denoted

$$|A| = |B|, \quad \text{if } f \text{ is onto}$$

if and only if there is a one-to-one correspondence (i.e., a bijection) from A to B . 若存在一个一一对应, 则他们是 $|A| = |B|$ onto

If there is a **one-to-one function (i.e., an injection)** from A to B , the cardinality of A is less than or the same as the cardinality of B and we write $|A| \leq |B|$.

When $|A| \leq |B|$ and A and B have different cardinality, we say that the cardinality of A is less than the cardinality of B and write $|A| < |B|$.

IMPORTANT!
重要!

Cardinality₂

Set是有限的、或是与正整数 $1, 2, 3, \dots$ onto $\rightarrow$ Countable!

Definition: A set that is either finite or has the same cardinality as the set of positive integers ($\mathbb{Z}^+$) is called **countable** (可數). A set that is not countable is **uncountable** (不可數).

The set of real numbers $\mathbb{R}$ is an uncountable set.

When an infinite set is countable (**countably infinite**) its cardinality is $\aleph_0$ (where $\aleph$ is aleph, the 1st letter of the Hebrew alphabet). We write $|S| = \aleph_0$ and say that **S has cardinality "aleph null."**



Showing that a Set is Countable

$a_1, a_2, a_3, a_4, \dots$
 ? ? ? ?

跟正整數($\mathbb{Z}^+$)一一对應!

An infinite set is countable if and only if it is possible to list the elements of the set in a sequence (indexed by the positive integers).

The reason for this is that a one-to-one correspondence f from the set of positive integers to a set S can be expressed in terms of a sequence $a_1, a_2, \dots, a_n, \dots$ where $a_1 = f(1)$, $a_2 = f(2)$, ..., $a_n = f(n)$, ...

Showing that a Set is Countable₁

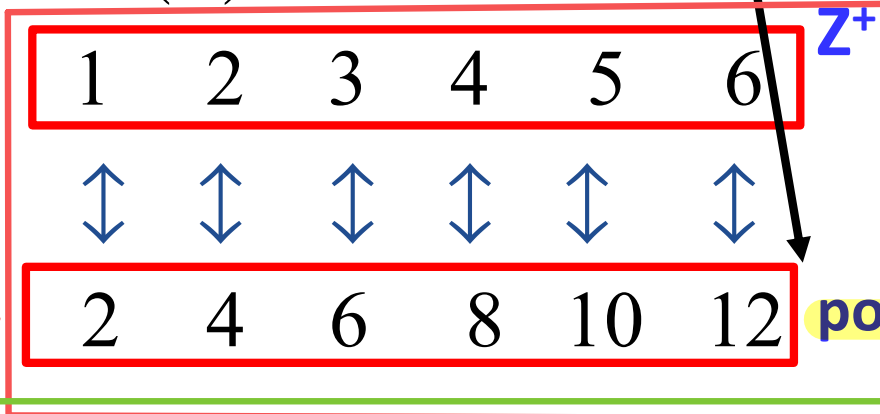
2, 4, 6, 8, ... Countable

Example 1: Show that the set of **positive even integers** E is countable set.

Solution: Let $f(x) = 2x$.

2STEP: 寫公式

Step 1.



0STEP 畫對應圖

Step 2. Show that f is a bijection

positive even integers

3STEP 考試要證明這種說明對應圖!

Then f is a **bijection** from $\mathbf{N}$ to E since f is both one-to-one and onto. To show that it is one-to-one, suppose that $f(n) = f(m)$. Then $2n = 2m$, and so $n = m$. To see that it is onto, suppose that t is an even positive integer. Then $t = 2k$ for some positive integer k and $f(k) = t$.

Showing that a Set is Countable₂

$\dots, -3, -2, -1, 0, 1, 2, 3, \dots$

Example 2: Show that the set of integers $\mathbf{Z}$ is countable.

Solution: Can list in a sequence:

$0, 1, -1, 2, -2, 3, -3, \dots$ 正確的擺法 →

☆

			①				
			$\mathbf{Z}^+$				
1	2	3	4	5	6	7	
↑	↑	↑	↑	↑	↑	↑	
0	1	-1	2	-2	3	-3	
			$\mathbf{Z}$				

Or can define a **bijection** from $\mathbf{N}$ to $\mathbf{Z}$:

- When n is even: $f(n) = n/2$
- When n is odd: $f(n) = -(n-1)/2$

③ (

$1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8$

X

錯的擺法

$0 \ 1 \ 2 \ 3 \ \dots \ -1$

The Positive Rational Numbers are Countable₁

Definition: A *rational number* (有理數) can be expressed as the ratio of two integers p and q such that $q \neq 0$.

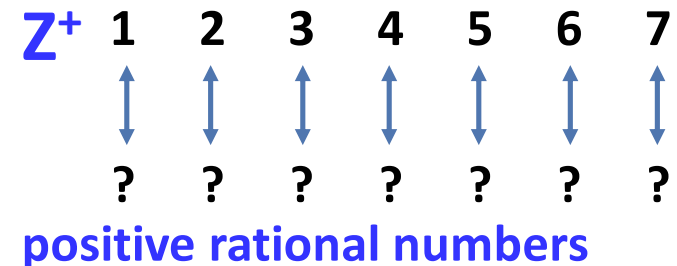
- $\frac{3}{4}$ is a rational number
- $\sqrt{2}$ is not a rational number.

證明正的有理數是 Countable

Example 3: Show that the positive rational numbers are countable.

Solution: The positive rational numbers are countable since they can be arranged in a sequence:

$r_1, r_2, r_3, \dots$



First list p/q with $p + q = 2$.

Next list p/q with $p + q = 3$

And so on.

1, $\frac{1}{2}$, 2, 3, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{2}{3}$, ...

First row $q = 1$.

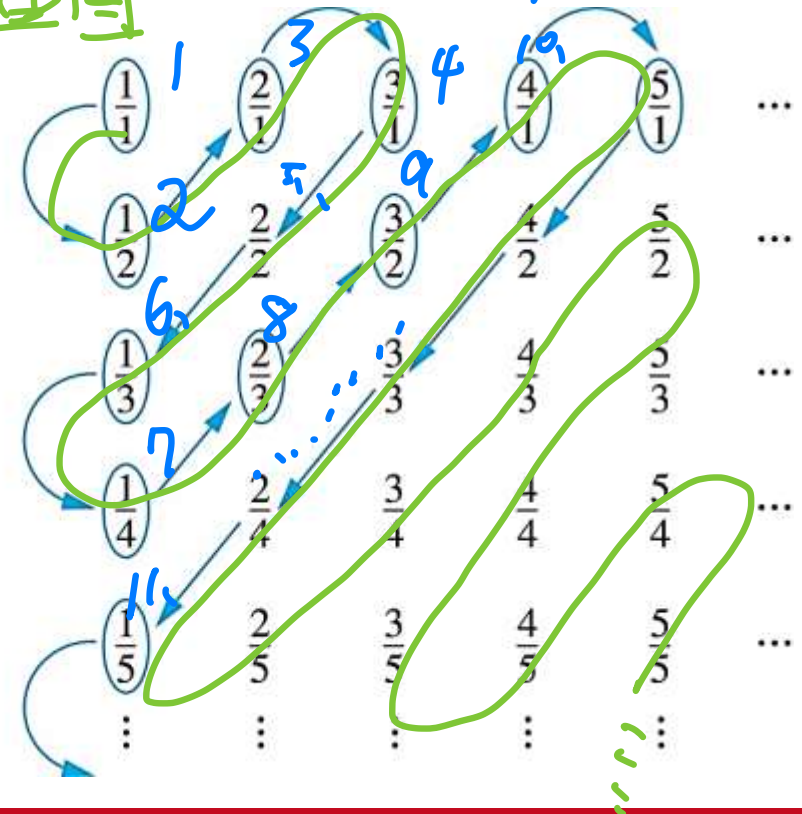
Second row $q = 2$.

etc.

→ $|-|$, on to

① 畫圖

Terms not circled
are not listed
because they
repeat previously
listed terms

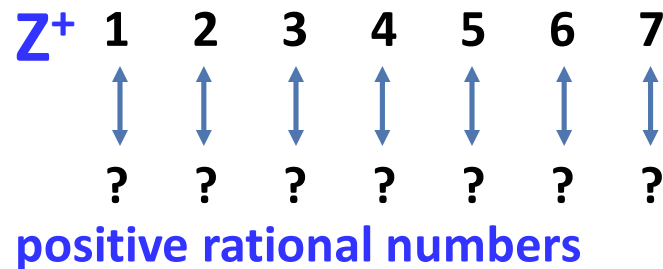


Strings₂

Example 4: Show that the set of **finite strings** S over a **finite alphabet** A is **countably infinite**.

Assume an alphabetical ordering of symbols in A

→ 假設有限!



✓
STEP 1!
夠嚴謹

1 2 3 4 27 28 29
 λ a b c z aa bb

Strings₂

Example 4: Show that the set of **finite strings** S over a finite alphabet A is **countably infinite**.

Assume an alphabetical ordering of symbols in A

Solution: Show that the strings can be listed in a sequence.
First list

1. All the strings of length 0 in alphabetical order.
2. Then all the strings of length 1 in lexicographic (as in a dictionary) order.
3. Then all the strings of length 2 in lexicographic order.
4. And so on.

This implies a bijection from $\mathbf{N}$ to S and hence it is a countably infinite set.

The set of all Java programs is countable.

Example 5: Show that the set of **all Java programs** is **countable**.

Solution: Let S be the set of strings constructed from the characters which can appear in a Java program. Use the ordering from the previous example. Take each string in turn:

- **Feed the string into a Java compiler.** (A Java compiler will determine if the input program is a syntactically correct Java program.)
- **If the compiler says YES, this is a syntactically correct Java program, we add the program to the list.**
- We move on to the next string.

In this way we construct an implied bijection from $\mathbf{N}$ to the set of Java programs. Hence, the set of Java programs is countable.

CLASSIC
經典題！

The Real Numbers are Uncountable

Example: Show that the set of real numbers is uncountable.

Solution: The method is called the Cantor diagonalization argument, and is a proof by contradiction.

- 1. Suppose R is countable. Then the real numbers between 0 and 1 are also countable (any subset of a countable set is countable - an exercise in the text).
- 2. The real numbers between 0 and 1 can be listed in order $r_1, r_2, r_3, \dots$.

3. Let the decimal representation of this listing be

4. Form a new real number with the decimal

expansion $r = 0.d_1d_2d_3d_4 \dots$

where

$$d_i = \begin{cases} 4 & \text{if } d_{ii} \neq 4 \\ 5 & \text{if } d_{ii} = 4 \end{cases}$$

For example,

$$\begin{aligned} r_1 &= 0.23794102 \dots \\ r_2 &= 0.44590138 \dots \\ r_3 &= 0.09118764 \dots \\ r_4 &= 0.80553900 \dots \\ r &= 0.4544\dots \end{aligned}$$

~~✗~~ $r_1 = 0.d_{11}d_{12}d_{13}d_{14}d_{15}d_{16}\dots$
 $r_2 = 0.d_{21}d_{22}d_{23}d_{24}d_{25}d_{26}\dots$
 $r_3 = 0.d_{31}d_{32}d_{33}d_{34}d_{35}$
 $r_4 = 0.d_{41}d_{42}d_{43}d_{44}d_{45}$

反駁: 0. d_{11} d_{22} d_{33} d_{44} d_{55}

↓ ↓ ↓ ↓ ↓
不跟 不跟 : : :
 d_{11} d_{22} : : :
一樣一樣

The Real Numbers are Uncountable

Example: Show that the set of real numbers is uncountable.

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For example,

$$r_1 = 0.23794102 \dots$$

$$r_2 = 0.44590138 \dots$$

$$r_3 = 0.09118764 \dots$$

$$r_4 = 0.80553900 \dots$$

$$r = 0.4544\dots$$

5. r is not equal to any of the $r_1, r_2, r_3, \dots$ Because it differs from r_i in its i th position after the decimal point. Therefore **there is a real number between 0 and 1 that is not on the list** since every real number has a unique decimal expansion. **Hence, all the real numbers between 0 and 1 cannot be listed**, so the set of real numbers between 0 and 1 is uncountable.
6. **Since a set with an uncountable subset is uncountable (an exercise), the set of real numbers is uncountable.**

Chapter 2.

Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

2.1 Sets

2.2 Set Operations

2.3 Functions

2.4 Sequences and Summations

2.5 Cardinality of Sets *< Countable or UnCountable Problem >*

2.6 Matrices 矩陣

Matrices

Matrices are useful discrete structures that can be used in many ways. For example, they are used to:

- describe certain types of functions known as linear transformations.
- Express which vertices of a graph are connected by edges (see Chapter 10).

In later chapters, we will see matrices used to build models of:

- **Transportation systems.**
- **Communication networks.**

Algorithms based on matrix models will be presented in later chapters.

Here we cover the aspect of **matrix arithmetic** that will be needed later.

Matrix

Definition: A *matrix* is a rectangular array of numbers. A matrix with m rows and n columns is called an $m \times n$ matrix.

- The plural of matrix is *matrices*.
- A matrix with the same number of rows as columns is called *square*.
- Two matrices are *equal* if they have the same number of rows and the same number of columns and the corresponding entries in every position are equal.

3 × 2 matrix

$$\begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 1 & 3 \end{bmatrix}$$

Columns 欄、列
↓
row → $\begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 1 & 3 \end{bmatrix}$

Notation

Let m and n be positive integers and let

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

The i th row of $\mathbf{A}$ is the $1 \times n$ matrix $[a_{i1}, a_{i2}, \dots, a_{in}]$. The j th column of $\mathbf{A}$ is the $m \times 1$ matrix:

$$\begin{bmatrix} a_{1j} \\ a_{2j} \\ \cdot \\ \cdot \\ a_{mj} \end{bmatrix}$$

The (i,j) th *element* or *entry* of $\mathbf{A}$ is the element a_{ij} . We can use $\mathbf{A} = [a_{ij}]$ to denote the matrix with its (i,j) th element equal to a_{ij} .

Matrix Arithmetic: Addition

Definition: Let $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$ be $m \times n$ matrices. The sum of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{A} + \mathbf{B}$, is the $m \times n$ matrix that has $a_{ij} + b_{ij}$ as its (i,j) th element. In other words, $\mathbf{A} + \mathbf{B} = [a_{ij} + b_{ij}]$.

Example:

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 2 & -3 \\ 3 & 4 & 0 \end{bmatrix} + \begin{bmatrix} 3 & 4 & -1 \\ 1 & -3 & 0 \\ -1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 4 & -2 \\ 3 & -1 & -3 \\ 2 & 5 & 2 \end{bmatrix}$$

Note that **matrices of different sizes can not be added.**

Matrix Multiplication

Definition: Let $\mathbf{A}$ be an $m \times k$ matrix and $\mathbf{B}$ be a $k \times n$ matrix. The *product* of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{AB}$, is the $m \times n$ matrix that has its (i,j) th element equal to the sum of the products of the corresponding elements from the i th row of $\mathbf{A}$ and the j th column of $\mathbf{B}$. In other words, if $\mathbf{AB} = [c_{ij}]$ then $c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ik}b_{kj}$.

Example:

$$\begin{bmatrix} 1 & 0 & 4 \\ 2 & 1 & 1 \\ 3 & 1 & 0 \\ 0 & 2 & 2 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 1 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 14 & 4 \\ 8 & 9 \\ 7 & 13 \\ 8 & 2 \end{bmatrix}$$

The product of two matrices is undefined when the number of columns in the first matrix is not the same as the number of rows in the second.

Illustration of Matrix Multiplication

The Product of $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ \mathbf{a_{i1}} & \mathbf{a_{i2}} & \dots & \mathbf{a_{ik}} \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ a_{m1} & a_{m2} & \dots & a_{mk} \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} b_{11} & a_{12} & \dots & \mathbf{b_{1j}} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & \mathbf{b_{2j}} & \dots & b_{2n} \\ \cdot & \cdot & & \cdot & & \cdot \\ \cdot & \cdot & & \cdot & & \cdot \\ b_{k1} & b_{k2} & \dots & \mathbf{b_{kj}} & \dots & b_{kn} \end{bmatrix}$$

$$\mathbf{AB} = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & b_{22} & \dots & c_{2n} \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & \mathbf{c_{ij}} & \cdot \\ \cdot & \cdot & & \cdot \\ c_{m1} & c_{m2} & \dots & c_{mn} \end{bmatrix}$$

$$\mathbf{c_{ij}} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ik}b_{kj}$$

Matrix Multiplication is not Commutative

Example: Let

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

Does **$AB = BA$** ?

① 對於任於矩陣,
 $AB=BA$ 一定成立?

Solution: 反例

$$AB = \begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix} \quad BA = \begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}$$

$AB \neq BA$

Identity Matrix and Powers of Matrices

Definition: The **identity matrix** (單位矩陣) of order n is the $n \times n$ matrix $\mathbf{I}_n = [\delta_{ij}]$, where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$.

$$\mathbf{I}_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & & \cdot \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

$$\mathbf{A}\mathbf{I}_n = \mathbf{I}_m\mathbf{A} = \mathbf{A}$$

when $\mathbf{A}$ is an $m \times n$ matrix

Powers of square matrices can be defined. When $\mathbf{A}$ is an $n \times n$ matrix, we have:

$$\mathbf{A}^0 = \mathbf{I}_n$$

$$\mathbf{A}^r = \underbrace{\mathbf{A}\mathbf{A}\mathbf{A}\dots\mathbf{A}}_{r \text{ times}}$$

Transposes of Matrices₁

轉置

Definition: Let $\mathbf{A} = [a_{ij}]$ be an $m \times n$ matrix. The **transpose** (轉置) of $\mathbf{A}$, denoted by $\mathbf{A}^t$, is the $n \times m$ matrix obtained by interchanging the rows and columns of $\mathbf{A}$.

If $\mathbf{A}^t = [b_{ij}]$, then $b_{ij} = a_{ji}$ for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$.

The transpose of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ is the matrix $\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

Transposes of Matrices₂

對稱

Definition: A square matrix $\mathbf{A}$ is called **symmetric** (對稱) if $\mathbf{A} = \mathbf{A}^t$. Thus $\mathbf{A} = [a_{ij}]$ is symmetric if $a_{ij} = a_{ji}$ for i and j with $1 \leq i \leq n$ and $1 \leq j \leq n$.

The matrix
$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$
 is symmetric.

Square matrices do not change when their rows and columns are interchanged.

Zero-One Matrices₁

Definition: A matrix all of whose entries are either 0 or 1 is called a **zero-one matrix**. (These will be used in Chapters 9 and 10.)

Algorithms operating on discrete structures represented by zero-one matrices are based on Boolean arithmetic defined by the following Boolean operations:

$$b_1 \wedge b_2 = \begin{cases} 1 & \text{if } b_1 = b_2 = 1 \\ 0 & \text{otherwise} \end{cases} \quad b_1 \vee b_2 = \begin{cases} 1 & \text{if } b_1 = 1 \text{ or } b_2 = 1 \\ 0 & \text{otherwise} \end{cases}$$

Zero-One Matrices₂

Definition: Let $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$ be an $m \times n$ **zero-one matrices**.

- The **join** of $\mathbf{A}$ and $\mathbf{B}$ is the zero-one matrix with (i,j) th entry $a_{ij} \vee b_{ij}$. The *join* of $\mathbf{A}$ and $\mathbf{B}$ is denoted by $\mathbf{A} \vee \mathbf{B}$.
- The **meet** of $\mathbf{A}$ and $\mathbf{B}$ is the zero-one matrix with (i,j) th entry $a_{ij} \wedge b_{ij}$. The *meet* of $\mathbf{A}$ and $\mathbf{B}$ is denoted by $\mathbf{A} \wedge \mathbf{B}$.

Joins and Meets of Zero-One Matrices

Example: Find the join and meet of the zero-one matrices

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix},$$

$$B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Solution: The **join** of **A** and **B** is

$$A \vee B = \begin{bmatrix} 1 \vee 0 & 0 \vee 1 & 1 \vee 0 \\ 0 \vee 1 & 1 \vee 1 & 0 \vee 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

The **meet** of **A** and **B** is

$$A \wedge B = \begin{bmatrix} 1 \wedge 0 & 0 \wedge 1 & 1 \wedge 0 \\ 0 \wedge 1 & 1 \wedge 1 & 0 \wedge 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Boolean Product of Zero-One Matrices₁

Definition: Let $\mathbf{A} = [a_{ij}]$ be an $m \times k$ zero-one matrix and $\mathbf{B} = [b_{ij}]$ be a $k \times n$ zero-one matrix. The **Boolean product** of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{A} \odot \mathbf{B}$, is the $m \times n$ zero-one matrix with (i,j) th entry

$$c_{ij} = (a_{i1} \wedge b_{1j}) \vee (a_{i2} \wedge b_{2j}) \vee \dots \vee (a_{ik} \wedge b_{kj}).$$

Example: Find the Boolean product of $\mathbf{A}$ and $\mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix},$$

$$\mathbf{B} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Boolean Product of Zero-One Matrices₂

Solution: The Boolean product $A \odot B$ is given by

$$A \odot B = \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \\ (0 \wedge 1) \vee (1 \wedge 0) & (0 \wedge 1) \vee (1 \wedge 1) & (0 \wedge 0) \vee (1 \wedge 1) \\ (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \end{bmatrix}$$

$$= \begin{bmatrix} 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \\ 0 \vee 0 & 0 \vee 1 & 0 \vee 1 \\ 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$A \times B > 1$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

Boolean Powers of Zero-One Matrices₁

連乘

Definition: Let $\mathbf{A}$ be a square zero-one matrix and let r be a positive integer. The **r th Boolean power** of $\mathbf{A}$ is the Boolean product of r factors of $\mathbf{A}$, denoted by $\mathbf{A}^{[r]}$. Hence,

$$\mathbf{A}^{[r]} = \underbrace{\mathbf{A} \odot \mathbf{A} \odot \cdots \odot \mathbf{A}}_{r \text{ times}}$$

We define $\mathbf{A}^{[0]}$ to be $\mathbf{I}_n$.

For example,

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}. \quad \mathbf{A}^{[1]} = \mathbf{A}^{[1]} \odot \mathbf{A}^{[0]} = \mathbf{A} \odot \mathbf{I}_n = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix} \odot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{A}^{[1]}$$

Boolean Powers of Zero-One Matrices₂

Example: Let

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Find $A^{[n]}$ for all positive integers n .

Solution:

$$A^{[2]} = A \odot A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$A^{[3]} = A^{[2]} \odot A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$A^{[4]} = A^{[3]} \odot A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$A^{[5]} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$A^{[n]} = A^{[5]}$ for all positive integers n with $n \geq 5$

10/31 期中 9:30 - 12:10

Homework Answers 會在兩天前 Release!